

Model B Comparison: OLD vs NEW Design

METIMMOX-1 Economic Evaluation

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1 Introduction

This document compares the OLD and NEW Model B (Focused) designs to document the methodological change and its impact on results.

1.1 Summary of Changes

Aspect	OLD Model B	NEW Model B
Biomarkers as main effects	crp + tlr + tmb_braf	crp + tmb_braf
Interaction terms	One per model	BOTH (crp:Rx + tmb_braf:Rx)
Strategies evaluated	control, crp, tlr, tmb_braf	control, crp, tmb_braf
Number of distinct formulas	3 (one per biomarker)	1 (shared formula)

1.2 Rationale for Change

The OLD Model B included TLR as a biomarker but evidence suggests TLR may not provide sufficient predictive value for treatment selection. The NEW Model B:

1. **Excludes TLR entirely** - focuses on CRP and TMB/BRAF only
2. **Includes both interaction terms** - providing estimates for how each biomarker modifies treatment effect while adjusting for the other

2 Methodology

2.1 Formula Definitions

2.1.1 OLD Model B Formulas

Three different formulas, each with one interaction term and all three biomarkers as main effects:

- CRP strategy: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tlr} + \text{tmb_braf} + \text{crp:Rx}$
- TLR strategy: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tlr} + \text{tmb_braf} + \text{tlr:Rx}$
- TMB/BRAF strategy: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tlr} + \text{tmb_braf} + \text{tmb_braf:Rx}$

2.1.2 NEW Model B Formula

One shared formula with both interaction terms (TLR excluded):

- All strategies: $\text{Surv}(\dots) \sim \text{Age} + \text{sex} + \text{Rx} + \text{crp} + \text{tmb_braf} + \text{crp:Rx} + \text{tmb_braf:Rx}$

3 Model Fitting

3.1 Model Fit Comparison (AIC)

Table 2: AIC Comparison: OLD vs NEW Model B

Model	OS AIC	PFS AIC
OLD B-CRP	620.1	451.1
OLD B-TLR	618.7	452.2
OLD B-TMB_BRAF	620.1	451.7
NEW B (shared)	630.7	481.8

4 Parameter Estimates Comparison

4.1 Overall Survival - All Coefficients

4.1.1 OLD B-CRP - OS

Term	Estimate	SE	HR (95% CI)
shape	1.201	0.182	3.325 (2.329-4.747)
rate	-3.179	0.589	0.042 (0.013-0.132)
Age	0.003	0.008	1.003 (0.987-1.019)
sex1	0.167	0.149	1.182 (0.883-1.582)
RxExperimental arm	-0.046	0.194	0.955 (0.653-1.398)
crp1	-0.429	0.314	0.651 (0.352-1.206)
tlr1	-0.573	0.168	0.564 (0.406-0.784)
tmb_braf1	-0.139	0.154	0.870 (0.644-1.176)
RxExperimental arm:crp1	0.136	0.380	1.146 (0.544-2.414)

4.1.2 OLD B-TLR - OS

Term	Estimate	SE	HR (95% CI)
shape	1.226	0.182	3.408 (2.386-4.869)
rate	-2.896	0.611	0.055 (0.017-0.183)
Age	0.002	0.008	1.002 (0.986-1.018)
sex1	0.145	0.148	1.156 (0.864-1.545)
RxExperimental arm	-0.268	0.256	0.765 (0.463-1.263)
crp1	-0.361	0.177	0.697 (0.493-0.986)
tlr1	-0.808	0.247	0.446 (0.275-0.722)
tmb_braf1	-0.139	0.148	0.870 (0.651-1.163)
RxExperimental arm:tlr1	0.402	0.316	1.495 (0.805-2.775)

4.1.3 OLD B-TMB - OS

Term	Estimate	SE	HR (95% CI)
shape	1.205	0.182	3.335 (2.336-4.761)
rate	-3.243	0.563	0.039 (0.013-0.118)
Age	0.003	0.008	1.003 (0.988-1.019)
sex1	0.170	0.148	1.185 (0.886-1.585)
RxExperimental arm	0.032	0.202	1.033 (0.696-1.534)
crp1	-0.319	0.182	0.727 (0.509-1.038)
tlr1	-0.569	0.167	0.566 (0.408-0.786)
tmb_braf1	-0.062	0.221	0.940 (0.609-1.449)
RxExperimental arm:tmb_braf1	-0.117	0.304	0.890 (0.491-1.614)

4.1.4 NEW B - OS

Term	Estimate	SE	HR (95% CI)
shape	1.058	0.181	2.881 (2.020-4.108)
rate	-3.605	0.623	0.027 (0.008-0.092)
Age	-0.001	0.009	0.999 (0.983-1.016)
sex1	0.202	0.158	1.224 (0.898-1.670)

RxExperimental arm	0.261	0.224	1.298 (0.836-2.016)
crp1	-0.556	0.338	0.574 (0.296-1.113)
tmb_braf1	-0.042	0.239	0.959 (0.600-1.532)
RxExperimental arm:crp1	0.029	0.408	1.029 (0.463-2.290)
RxExperimental arm:tmb_braf1	-0.111	0.325	0.895 (0.474-1.691)

4.2 Progression-Free Survival - All Coefficients

4.2.1 OLD B-CRP - PFS

Term	Estimate	SE	HR (95% CI)
shape	1.215	0.190	3.370 (2.324-4.888)
rate	-1.680	0.597	0.186 (0.058-0.600)
Age	-0.001	0.009	0.999 (0.982-1.016)
sex1	0.115	0.165	1.122 (0.813-1.550)
RxExperimental arm	0.169	0.197	1.184 (0.805-1.742)
crp1	0.014	0.420	1.014 (0.445-2.311)
tlr1	-1.114	0.168	0.328 (0.236-0.456)
tmb_braf1	-0.323	0.174	0.724 (0.514-1.018)
RxExperimental arm:crp1	-0.704	0.485	0.494 (0.191-1.280)

4.2.2 OLD B-TLR - PFS

Term	Estimate	SE	HR (95% CI)
shape	1.194	0.190	3.299 (2.273-4.788)
rate	-1.711	0.634	0.181 (0.052-0.626)
Age	-0.002	0.009	0.998 (0.981-1.015)
sex1	0.128	0.170	1.137 (0.815-1.585)
RxExperimental arm	0.237	0.285	1.267 (0.725-2.215)
crp1	-0.527	0.206	0.590 (0.394-0.885)
tlr1	-0.928	0.273	0.395 (0.232-0.675)
tmb_braf1	-0.387	0.169	0.679 (0.488-0.947)
RxExperimental arm:tlr1	-0.295	0.349	0.745 (0.376-1.476)

4.2.3 OLD B-TMB - PFS

Term	Estimate	SE	HR (95% CI)
shape	1.197	0.190	3.311 (2.283-4.804)
rate	-1.556	0.576	0.211 (0.068-0.653)
Age	-0.003	0.008	0.997 (0.980-1.013)
sex1	0.097	0.167	1.102 (0.795-1.528)
RxExperimental arm	0.181	0.218	1.198 (0.782-1.837)
crp1	-0.458	0.222	0.633 (0.409-0.978)
tlr1	-1.105	0.168	0.331 (0.238-0.460)
tmb_braf1	-0.191	0.250	0.826 (0.506-1.349)
RxExperimental arm:tmb_braf1	-0.388	0.347	0.678 (0.343-1.340)

4.2.4 NEW B - PFS

Term	Estimate	SE	HR (95% CI)
shape	0.707	0.181	2.027 (1.421-2.893)
rate	-2.507	0.773	0.082 (0.018-0.371)
Age	-0.011	0.011	0.989 (0.968-1.011)
sex1	0.139	0.210	1.149 (0.762-1.734)
RxExperimental arm	0.560	0.296	1.751 (0.980-3.131)
crp1	-0.263	0.542	0.769 (0.266-2.222)
tmb_braf1	-0.178	0.337	0.837 (0.432-1.622)

RxExperimental arm:crp1	-0.661	0.620	0.516 (0.153-1.741)
RxExperimental arm:tmb_braf1	-0.389	0.449	0.678 (0.281-1.634)

5 Survival Curves Comparison

5.1 Overall Survival Curves

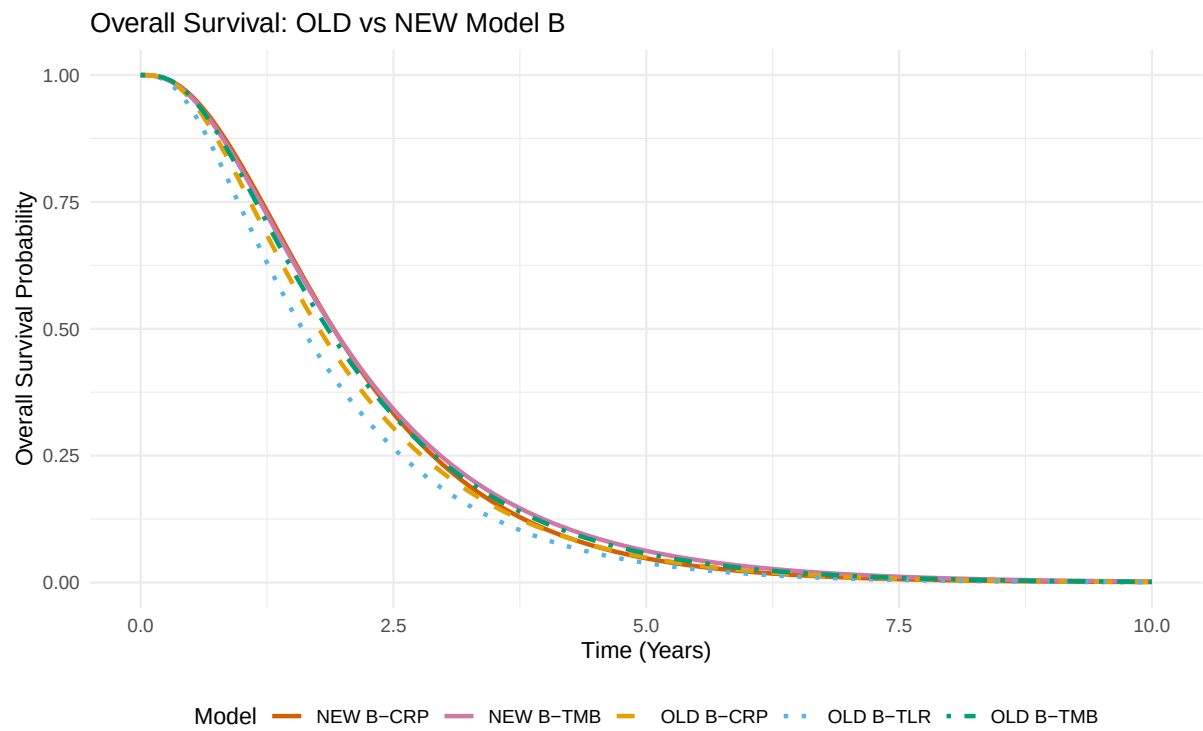


Figure 1: Overall Survival Curves: OLD vs NEW Model B (population-weighted)

5.2 Progression-Free Survival Curves

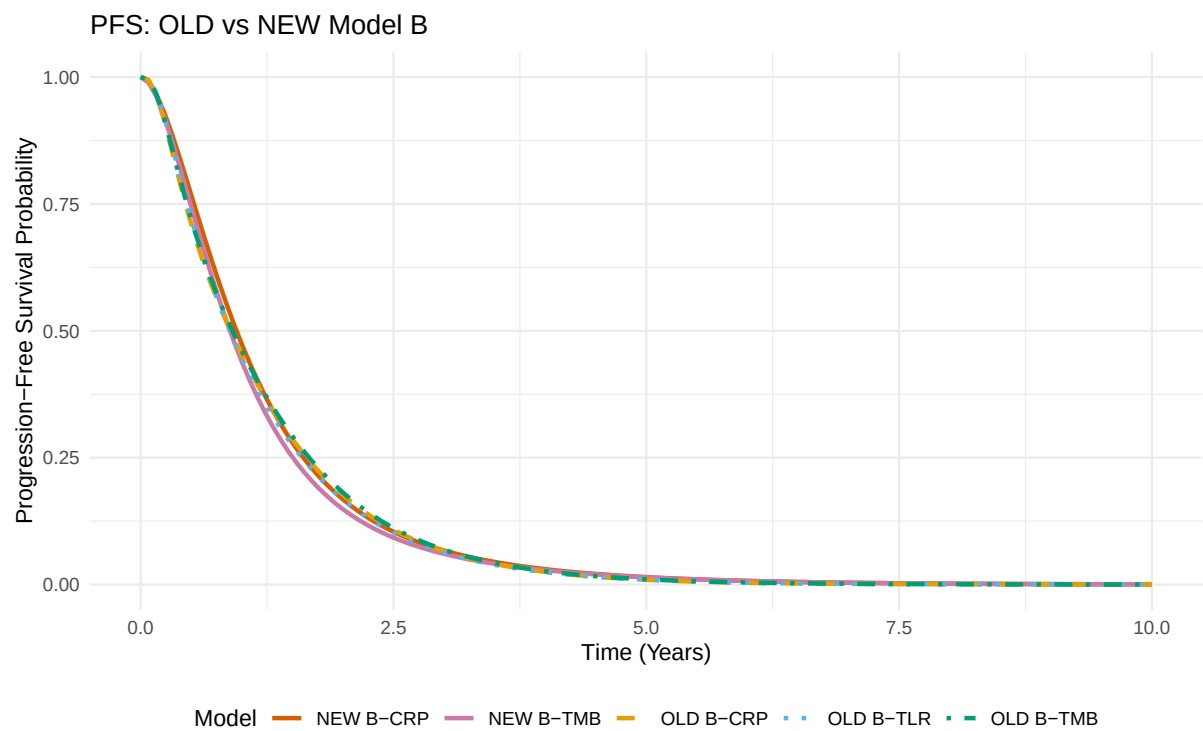


Figure 2: Progression-Free Survival Curves: OLD vs NEW Model B (population-weighted)

6 Life Years, QALYs, and Costs

6.1 Discounted Life Years Comparison

Table 11: Discounted Life Years Comparison (dr = 4%)

Outcome	Control	TLR (OLD)	CRP Strategy			TMB/BRAF Strategy		
			OLD	NEW	Delta	OLD	NEW	Delta
LY (OS)	2.044	1.849	2.001	2.087	0.086	2.087	2.138	0.051
LY (PFS)	1.165	1.144	1.156	1.199	0.043	1.175	1.136	-0.038
LY (Progressed)	0.879	0.705	0.846	0.888	0.043	0.913	1.002	0.089

6.2 Undiscounted Life Years Comparison

Table 12: Undiscounted Life Years Comparison

Outcome	Control	TLR (OLD)	CRP Strategy			TMB/BRAF Strategy		
			OLD	NEW	Delta	OLD	NEW	Delta
LY (OS)	2.173	1.956	2.124	2.216	0.091	2.221	2.279	0.058
LY (PFS)	1.214	1.190	1.204	1.251	0.047	1.224	1.185	-0.040
LY (Progressed)	0.959	0.765	0.921	0.965	0.044	0.996	1.094	0.097

6.3 QALYs Comparison

Table 13: Discounted QALYs Comparison (Utilities: PF=0.9077, P=0.9005)

Metric	Control	TLR (OLD)	CRP Strategy			TMB/BRAF Strategy		
			OLD	NEW	Delta	OLD	NEW	Delta
QALYs	1.849	1.673	1.81	1.888	0.078	1.888	1.933	0.045

6.4 Costs Comparison

Table 14: Treatment Costs Comparison

Metric	Control	TLR (OLD)	CRP Strategy			TMB/BRAF Strategy		
			OLD	NEW	Delta	OLD	NEW	Delta
Cost (EUR)	6,832	74,935	43,375	43,375	0	55,002	55,002	0

6.5 ICER Comparison - OLD Model B

Table 15: Cost-Effectiveness: OLD Model B (vs Control)

Strategy	Cost (EUR)	QALYs	Incr. Cost	Incr. QALYs	ICER
Control	6,832	1.849	-	-	-
CRP	43,375	1.810	36,543	-0.039	-
TMB/BRAF	55,002	1.888	48,170	0.039	1,242,002

6.6 ICER Comparison - NEW Model B

Table 16: Cost-Effectiveness: NEW Model B (vs Control)

Strategy	Cost (EUR)	QALYs	Incr. Cost	Incr. QALYs	ICER
Control	6,832	1.849	-	-	-
CRP	43,375	1.888	36,543	0.039	939,142
TMB/BRAF	55,002	1.933	48,170	0.084	572,263

6.7 ICER Change: OLD vs NEW

Table 17: ICER Change: OLD vs NEW Model B (EUR/QALY)

Strategy	OLD ICER	NEW ICER	Delta	Change
CRP	-	939,142	-	-
TMB/BRAF	1,242,002	572,263	-669,739	-53.9\%

7 Summary

The redesign of Model B from OLD (all biomarkers as main effects with single interaction per strategy) to NEW (CRP + TMB/BRAF only with both interactions) has notable implications for cost-effectiveness results.

7.1 Key Findings

CRP Strategy: Under the OLD Model B, the CRP strategy showed negative or negligible incremental QALYs compared to control (-0.039 QALYs), which resulted in no valid ICER (dominated or equivalent). With the NEW Model B, the CRP strategy now shows a positive incremental effect (0.039 QALYs), yielding a calculable ICER of EUR 939,142.

TMB/BRAF Strategy: The effect of the TMB/BRAF strategy is larger under the NEW model. OLD Model B showed incremental QALYs of 0.039 (ICER: EUR 1,242,002), while NEW Model B shows 0.084 incremental QALYs (ICER: EUR 572,263). This represents a larger treatment benefit under the revised model specification.

7.2 Implications

The NEW Model B design has the following implications:

1. **TLR is excluded:** The analysis now focuses on only two biomarker strategies (CRP and TMB/BRAF) rather than three.
2. **Both interactions estimated simultaneously:** The shared formula includes both `crp:Rx` and `tmb_braf:Rx` interaction terms, meaning each biomarker's treatment effect is estimated while adjusting for the other.
3. **Changed cost-effectiveness profile:** The model redesign affects the estimated incremental effects and ICERs, with CRP showing improved cost-effectiveness and TMB/BRAF showing a larger treatment benefit (and consequently lower ICER) compared to the OLD model specification.

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